Roll No.: DL.AL.U4AID24135

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Amrita School of Artificial Intelligence, Faridabad

B.Tech. AI/DS Mid-Term Examinations – October 2025

First Semester

Artificial Intelligence & Data Science

23AID201 Simulation, Modelling and Analysis

Duration: Two hours

Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Apply various modeling techniques including physical, mathematical, and computer-based modeling for engineering applications.
CO02	Analyze different system models, including basic ones such as mechanical, electrical, hydraulic, pneumatic, and thermal systems, and advanced ones such as electro-mechanical, hydromechanical, and robotic systems
CO03	Apply various methods for modeling and simulating the behavior dynamic systems, including bond graph modeling, simulation using MATLAB, and parameter estimation methods.
CO04	Design simulations and analysis for engineering problems using optimization, block diagram algebra, signal flow graphs, state variable formulation, frequency response, and Bode plot.

Answer all questions

Part A

1) Define a model. What does it mean when we say a model is (i) explicative (ii) predictive? Give an example to illustrate.

[2] [CO01][BTL 1]

2) We are given two systems: a mass-spring-damper system described by $m\ddot{x} + c\dot{x} + kx = \eta(t)$, with $\eta(t)$ being random vibrations, c being damping constant, m being mass and k being stiffness of the spring, and a pendulum system described by $mL\ddot{\theta} + mg \sin\theta = 0$. Classify both these systems into whether they are linear or non-linear and deterministic or stochastic.

[2] [CO01][BTL 3]

3) What are variables and relations in a mathematical model? Give any two kinds of each.

[2] [CO02][BTL 1]

4) What is a dynamic system? Give the general mathematical representation. What are the sources of dynamics and examples of dynamic systems.

[2] [CO02][BTL 2]

5) Find the equilibrium points of the following dynamic systems:

$$\dot{x} = x(3 - 2x - y), \dot{y} = y(2 - x - y)$$

Determine the form of the Jacobian at these points.

[2] [CO02][BTL 4]

Part B

6) Given $F(t) = \frac{t}{t+0.6}$ as the proportion of nectar a bee consumes in t seconds at a flower and τ as the travel time between flowers, determine the time the bee should spend at each flower to maximize its overall nectar collection rate, where time spent at each flower includes the average travel time between flowers.

[5] [CO02][BTL 6]

7) Give the method of problem and model formulation, using the illustrative example of projectile motion in classical mechanics.

[5] [CO01][BTL 2]

8) Evaluate the solutions for $\dot{x} = -2x, x(0) = 1$ using the analytical method and one numerical method, for time-steps $t = 0.2, 0.4, 0.6, 0.8$ and 1.

[5] [CO01][BTL 5]

9) Determine the kinds of stability associated with Lyapunov analysis. Check whether the system

$$\dot{x} = -3x, x(0) = 2$$

is Lyapunov and asymptotically stable at the equilibrium point.

[5] [CO02][BTL 4]

Part C

10) The Lotka–Volterra predator–prey model is described by the following system of differential equations:

$$\dot{x} = \alpha x - \beta xy,\ \dot{y} = \delta xy - \gamma y, \alpha > 0, \beta > 0, \gamma > 0, \delta > 0$$

If $\alpha = 0.1, \beta = 0.02, \gamma = 0.3, \delta = 0.01, x(0) = 40, y(0) = 9$, then determine the equilibrium points and the Jacobian matrix at these points, for this system.

[5] [CO01][BTL 3]

Analyze the stability of the equilibrium point using the Jacobian.

[3] [CO01][BTL 4]

Using Euler's method with a time step $\Delta t = 0.5$, compute approximate values of x and y at $t =$ 0.5,1.0.

[2] [CO01][BTL 5]

11) Define and classify sources and passive elements in bond-graph modelling, with their associated mathematical expressions.

[5] [CO02][BTL 1]

Construct the bond graph model for a mechanical system that consists of a mass $m = 2$ kg, lying on a spring with stiffness $k = 50$ N/m, connected in series with a damper with damping coefficient $b = 10$ Ns/m, which is fixed to the ground. The system is driven by an external force $F(t)$.

[5] [CO02][BTL 6]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	24	BTL 1	9
CO02	26	BTL 2	7
CO03	0	BTL 3	7
CO04	0	BTL 4	10
		BTL 5	7
		BTL 6	10

Bloom's Taxonomy Levels (attached for reference)

Level 1 – Remember

Level 2 - Understand

Level 3 – Apply

Level 4 - Analyze

Level 5 - Evaluate

Level 6- Create